Title: Monte Carlo Simulation for Portfolio Risk and Stress Testing

1. Introduction

In modern financial markets, accurate modelling of risk is essential for portfolio management and regulatory compliance. Traditional models often assume normally distributed returns, which fail to capture extreme events observed in practice.

This study develops a Monte Carlo-based simulation framework to model portfolio returns under different distributional assumptions, including Gaussian, mixture distributions, and jump-diffusion processes.

2. Methodology

2.1 Correlated Asset Simulation

Asset returns were simulated using a multivariate normal framework. Correlation between assets was imposed using the Cholesky decomposition:

$$\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^{\boldsymbol{T}}$$

Independent normal variables were transformed via:

$$\boldsymbol{X} = \boldsymbol{L}\boldsymbol{Z}$$

This ensures the simulated returns match the target covariance structure.

2.2 Models Considered

(i) Gaussian Model

Assumes returns follow a normal distribution:

- symmetric
- thin tails

(ii) Gaussian Mixture Model

Models returns as a weighted combination of distributions:

- captures skewness
- incorporates stress regimes

(iii) Merton Jump-Diffusion

Extends GBM by adding jumps:

$$dS = \mu S dt + \sigma S dW + J_t$$

This captures:

- sudden market crashes
- discontinuities

2.3 Risk Metrics

Value at Risk (VaR)

$$VaR_\alpha = \text{quantile at } (1 - \alpha)$$

Expected Shortfall (ES)

$$ES = E[X \mid X < VaR]$$

3. Results

3.1 Distribution Analysis (Mixture Model)

- Mean: -0.00408
- Standard deviation: 0.02191
- Skewness: -2.80
- Kurtosis: 10.75

These results indicate:

- strong downside asymmetry
- significant tail risk

3.2 Correlation Validation

Simulated correlations closely matched theoretical values, confirming correct implementation of Cholesky decomposition.

3.3 Monte Carlo Validation (GBM)

- Simulated mean: 51.94
- Theoretical mean: 52.04

This demonstrates convergence of Monte Carlo estimation.

3.4 Risk Comparison

Model	VaR (95%)	ES
Normal	-0.0330	-0.0413
Mixture	-0.0499	-0.0814
Jump	-0.4264	-0.5339

4. Discussion

The results highlight significant differences in risk estimation across models:

- The Gaussian model underestimates extreme losses due to its thin tails.
- The mixture model introduces skewness and captures more realistic market behavior.
- The jump-diffusion model produces extreme tail events, reflecting real-world crises.

This demonstrates that model choice has a critical impact on risk assessment.

5. Conclusion

Monte Carlo simulation provides a flexible framework for modelling financial risk.
Incorporating realistic features such as fat tails and jumps significantly improves the accuracy of risk estimates.

The study confirms that:

- normal assumptions are insufficient
- tail-aware models are essential
- correlation plays a critical role

6. Recommendations

Future work should include:

- calibration to real market data
- back testing of VaR predictions
- inclusion of stochastic volatility models